Functional Annotation Report: *pyrB* (PP_4998, UniProt Q88D30) — Aspartate Carbamoyltransferase Catalytic Subunit of *Pseudomonas putida* KT2440

Target: UniProt **Q88D30** | Gene **pyrB** / OrderedLocusName **PP_4998** | EC **2.1.3.2** **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950 / KT2440) **Date:** 2026-09-01

Summary

The gene **pyrB** (ordered locus name **PP_4998**; UniProt accession **Q88D30**) of *Pseudomonas putida* strain KT2440 encodes the **catalytic subunit of aspartate carbamoyltransferase** (aspartate transcarbamoylase; **ATCase**; EC 2.1.3.2). ATCase catalyzes the **first committed and rate-limiting step of the de novo pyrimidine biosynthetic pathway**: the transfer of a carbamoyl group from **carbamoyl phosphate** to the α-amino group of **L-aspartate**, producing **N-carbamoyl-L-aspartate** and **inorganic phosphate**. This reaction commits the cell's metabolic flux toward uridine-5′-monophosphate (UMP), the precursor from which all other pyrimidine nucleotides are derived. The enzyme functions in the **cytoplasm** as a soluble protein.

The identity of Q88D30 as a *pyrB*/ATCase catalytic subunit is strongly supported by convergent lines of evidence: (1) UniProt's HAMAP rule-based family and domain annotation places it firmly in the aspartate/ornithine carbamoyltransferase family with all five diagnostic InterPro domains; (2) direct sequence analysis of the 334-residue Q88D30 protein reveals intact, positionally conserved catalytic motifs (the carbamoyl-phosphate-binding "STRTR" loop and the aspartate-binding "HPTQ" motif); (3) global alignment against the extensively characterized *Escherichia coli* catalytic subunit shows ~39% identity with exact conservation of the catalytic carbamoyl-phosphate-binding arginine; and (4) the KT2440 genome confirms the experimentally described *pyrB–pyrC′* gene arrangement.

A distinctive and important feature of the *P. putida* enzyme, established by direct biochemical and genetic characterization, is that its **quaternary structure differs fundamentally from the textbook *E. coli* ATCase**. Rather than the *E. coli* class-C architecture of six catalytic (PyrB) plus six regulatory (PyrI) chains, the *Pseudomonas*-type enzyme is a **dodecamer of six catalytic PyrB subunits and six catalytically inactive dihydroorotase-homolog PyrC′ subunits** (encoded by the adjacent PP_4999). *P. putida* possesses **no separate PyrI regulatory subunit**; instead, the regulatory nucleotide-binding site resides in a **unique N-terminal extension of PyrB itself**, and the inactive PyrC′ serves a purely structural role, maintaining the dodecameric assembly required for activity. Consequently, UniProt's rule-propagated "$2C_3:3R_2$ heterododecamer with six PyrI chains" subunit annotation is **inaccurate for this organism**, while its FUNCTION, CATALYTIC ACTIVITY, and PATHWAY annotations remain correct.

Key Findings

Finding 1 — *pyrB* (Q88D30) encodes the catalytic subunit of aspartate carbamoyltransferase (ATCase, EC 2.1.3.2)

UniProt Q88D30 annotates PP_4998/*pyrB* as the ATCase catalytic subunit (EC 2.1.3.2), a member of the aspartate/ornithine carbamoyltransferase family, carrying the diagnostic **Asp/Orn carbamoyltransferase P-binding domain (IPR006132)** and **Asp/Orn-binding domain (IPR006131)**, along with IPR006130, IPR036901, and IPR002082. The enzyme catalyzes:

> **carbamoyl phosphate + L-aspartate → N-carbamoyl-L-aspartate + phosphate + H^+**

The chemical mechanism has been established rigorously in the *E. coli* ortholog using ^{13}C and ^{15}N kinetic isotope effect studies. The reaction proceeds by an **ordered nucleophilic attack of the aspartate α-amino group on the carbonyl carbon of carbamoyl phosphate**, forming a **tetrahedral intermediate** that collapses, following an intramolecular proton transfer, into the products N-carbamoyl-L-aspartate and inorganic phosphate. As reported by Waldrop and colleagues: *"Nucleophilic attack on the carbonyl carbon of carbamyl phosphate by the alpha-amino group of L-aspartate results in the formation of a tetrahedral intermediate. An intramolecular proton transfer leads to formation of products N-carbamyl-L-aspartate and inorganic phosphate"* (PMID: 1633169).

The pathway role of this reaction is definitive: ATCase catalyzes **the committed step of de novo pyrimidine biosynthesis**. As stated in a study of the plant enzyme: *"Aspartate transcarbamoylase (ATCase, EC 2.1.3.2) catalyzes the committed step in the de novo synthesis of uridine-5'-monophosphate (UMP), from which all other pyrimidine nucleotides are made"* (PMID: 18053734). Because this step commits metabolic resources irreversibly toward pyrimidines, it is a natural and universal locus of metabolic regulation.

Finding 2 — The *P. putida* ATCase is a dodecameric $PyrB_6$/$PyrC'_6$ complex with an internal regulatory site, not the *E. coli* c_6r_6 architecture

The most mechanistically distinctive feature of this enzyme comes from direct sequencing and biochemical characterization of the *P. putida pyrBC′* locus by Schurr and colleagues. The *pyrB* gene is **1,005 bp long and encodes the 334-amino-acid, 36.4-kDa catalytic subunit**. Immediately downstream, an overlapping and co-translated gene encodes a **424-residue, 44.2-kDa polypeptide homologous to dihydroorotase (DHOase)** but which is **catalytically inactive** — it lacks the conserved catalytic histidyl residues and does not complement *E. coli pyrC* auxotrophs.

Crucially, the native *P. putida* ATCase *"does not possess dissociable regulatory and catalytic functions but instead apparently contains the regulatory nucleotide binding site within a unique N-terminal extension of the pyrB-encoded subunit. The first gene, pyrB, is 1,005 bp long and encodes the 334-amino-acid, 36.4-kDa catalytic subunit of the enzyme"* (PMID: 7896697). This means that, unlike *E. coli* ATCase (which separates into distinct catalytic and regulatory subunits), the *Pseudomonas* enzyme integrates its allosteric regulatory site into the catalytic polypeptide itself.

The inactive DHOase homolog (designated **PyrC′**) is not a functional dihydroorotase but a **structural partner**: *"The proposed function for the vestigial DHOase is to maintain ATCase activity by conserving the dodecameric assembly of the native enzyme"* (PMID: 7896697). The holoenzyme is therefore a **dodecamer of six active PyrB and six inactive PyrC′ chains**, and PyrC′ is required to hold the assembly together and thereby maintain catalytic activity.

This *Pseudomonas*-type ("class A") architecture is consistent with the known structural diversity of the transcarbamoylase family. For example, the related catabolic ornithine transcarbamoylase of *Pseudomonas aeruginosa* is *"a dodecamer composed of four trimers organized in a tetrahedral manner"* with 23 point-group symmetry, in contrast to the pseudo-32 symmetry of *E. coli* ATCase (PMID: 7479879).

Finding 3 — ATCase uses an ordered kinetic mechanism (carbamoyl phosphate binds first) and is a cytoplasmic pyrimidine-pathway enzyme

The catalytic cycle proceeds through an **ordered-binding kinetic mechanism** in which **carbamoyl phosphate (CP) binds first**, forming an enzyme–CP complex before L-aspartate binds. As described for the transcarbamoylase family: *"Both of these transcarbamoylases use an ordered-binding mechanism in which CP binds first, allowing the formation of an enzyme"* complex (PMID: 18971327). This ordered binding has a physiological benefit beyond kinetics: CP binding **stabilizes the otherwise thermally labile intermediate**, protecting it from decomposition.

The active site is formed at the interface between adjacent subunits of the catalytic trimer, and its key residues are conserved across the family. In the *E. coli* catalytic chain, **Arg54 interacts with both the anhydride oxygen and a phosphate oxygen of carbamoyl phosphate** (PMID: 1303763), positioning the substrate for catalysis; **Arg105 and Leu267** help orient the aspartate and CP substrates (PMID: 17603076); and **Lys84 from an adjacent chain** (part of the "80s loop") is critical for catalysis and cooperativity (PMID: 10386880). The reaction occurs in the **cytoplasm**, consistent with the soluble, non-membrane character of the pyrimidine biosynthetic enzymes.

Finding 4 — Sequence analysis of Q88D30 confirms intact ATCase catalytic signatures and the *Pseudomonas* N-terminal extension

Direct examination of the Q88D30 primary sequence independently corroborates the annotation. The protein is **exactly 334 residues long**, matching the catalytic subunit size reported experimentally: *"The first gene, pyrB, is 1,005 bp long and encodes the 334-amino-acid, 36.4-kDa catalytic subunit of the enzyme"* (PMID: 7896697). The correspondence of the retrieved sequence length with the experimentally determined subunit confirms correct gene-product identity.

The sequence contains the **diagnostic carbamoyl-phosphate-binding loop motif "STRTR"** at positions 69–73 (in the context …FFEN**STRTR**TTF…), whose central arginine corresponds to the catalytic **Arg54** of *E. coli* ATCase that contacts carbamoyl phosphate (PMID: 1303763). It also carries the conserved second-domain **"HPTQ" motif** at positions 151–154 (…NGGDGRHA**HPTQ**GML…), characteristic of the aspartate-binding domain. Notably, the catalytic P-loop motif appears **~15–17 residues later** than in the shorter *E. coli* PyrB (where the equivalent arginine is near residue 52–54), consistent with the **unique N-terminal extension** that in *Pseudomonas* houses the regulatory nucleotide-binding site.

Finding 5 — Q88D30 is a bona fide ortholog of the *E. coli* ATCase catalytic subunit (~39% identity, catalytic motif exactly conserved)

A global Needleman–Wunsch alignment of *P. putida* PyrB (Q88D30, 334 aa) against the extensively characterized *E. coli* K-12 PyrB catalytic subunit (P0A786, 311 aa) yields **117/303 = 38.6% identity** over the aligned core. The carbamoyl-phosphate-binding catalytic loop aligns **exactly** (*P. putida* FEN-**STRTR**-TTF vs. *E. coli* FEA-**STRTR**-LSF), confirming positional conservation of the catalytic arginine (*E. coli* Arg54). The ~23-residue length difference is fully accounted for by the *Pseudomonas* N-terminal regulatory extension.

This orthology assignment is reinforced by the recognized sequence grouping of *Pseudomonas* ATCases. In a study of the *Thermus* strain ZO5 enzyme: *"The deduced amino acid sequence of Thermus strain ZO5 aspartate carbamoyltransferase (ATCase; encoded by pyrB) exhibits the highest similarities (about 50% identical amino acids) with ATCases from Pseudomonas sp."* (PMID: 9171389). This confirms that *Pseudomonas* ATCases form a recognizable, distinct sequence group within the family.

The same study documents **pyrimidine end-product transcriptional control**: *"In Thermus strain ZO5, pyrB and pyrC gene expression is repressed three- to fourfold by uracil and increased twofold by arginine"* (PMID: 9171389). This indicates that, in addition to allosteric control at the protein level, *pyr* genes including *pyrB* are subject to gene-expression-level regulation by pyrimidine availability — a layer of control expected to operate on the *P. putida pyr* regulon as well.

Finding 6 — KT2440 genome confirms the *pyrB*(PP_4998)–*pyrC′*(PP_4999) locus and lacks *pyrI*; UniProt's *E. coli*-style subunit annotation is an inaccurate rule propagation

In the *P. putida* KT2440 reference genome, *pyrB* (PP_4998, Q88D30, 334 aa, 36.3 kDa) is immediately adjacent to **PP_4999 (Q88D29, 423 aa, 44.1 kDa)**, a dihydroorotase-like protein. This matches the 424-aa/44.2-kDa inactive DHOase homolog (PyrC′) described biochemically by Schurr et al. (PMID: 7896697). A **separate, genuine dihydroorotase, *pyrC* (PP_1086, Q88NW7, 348 aa, EC 3.5.2.3)**, is located elsewhere in the genome and provides the actual dihydroorotase step (step 3) of the pyrimidine pathway. This genomic arrangement — an inactive DHOase homolog fused into the ATCase operon plus a genuine DHOase elsewhere — is exactly what the *Pseudomonas*-type ATCase model predicts.

A UniProt query for a *pyrI* (ATCase regulatory subunit) gene in *P. putida* KT2440 (taxid 160488) returns **no hits** — the genome encodes no PyrI. This directly supports the biochemical conclusion that *"The P. putida ATCase does not possess dissociable regulatory and catalytic functions but instead apparently contains the regulatory nucleotide binding site within a unique N-terminal extension of the pyrB-encoded subunit"* (PMID: 7896697).

Nonetheless, UniProt's **SUBUNIT** field for Q88D30 (a rule-based annotation, ECO:0000255, HAMAP-Rule MF_00001) states: *"Heterododecamer (2C3:3R2) of six catalytic PyrB chains … and six regulatory PyrI chains"* — the *E. coli* class-C architecture. **This is an inaccurate propagation of a general rule to an organism whose enzyme has a genuinely different quaternary structure.** By contrast, UniProt's curated **FUNCTION**, **CATALYTIC ACTIVITY** (carbamoyl phosphate + L-aspartate = N-carbamoyl-L-aspartate + phosphate + H^+), and **PATHWAY** (UMP biosynthesis de novo, step 2 of 3) annotations are accurate.

Mechanistic Model and Interpretation

The catalyzed reaction and its pathway context

ATCase performs the **second enzymatic step and the first committed step** of de novo pyrimidine biosynthesis. The pathway from central metabolism to UMP proceeds as follows:

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Glutamine + 2 ATP + HCO3-
         │  (CPSase, PyrA)
         ▼
Carbamoyl phosphate  ──────┐
                           │  + L-aspartate
                           ▼
     ╔═══════════════════════════════╗
     ║  ATCase  (pyrB / PP_4998)     ║  ← COMMITTED STEP
     ║  EC 2.1.3.2                   ║
     ╚═══════════════════════════════╝
                           │
                           ▼
           N-carbamoyl-L-aspartate + Pi
                           │  (DHOase, genuine pyrC / PP_1086)
                           ▼
                 Dihydroorotate
                           │  (DHODH, pyrD)
                           ▼
                     Orotate
                           │  (OPRTase, pyrE)
                           ▼
                     Orotidine-5'-P
                           │  (OMP decarboxylase, pyrF)
                           ▼
                       U M P  ──►  all other pyrimidine nucleotides
```

The enzyme carries out its reaction in the **cytoplasm**. Its high specificity is for **L-aspartate** as the amino-group acceptor (the related family member ornithine transcarbamoylase uses L-ornithine instead; the two enzymes share the CP-binding domain but diverge in the second, substrate-binding domain). ATCase can weakly carbamoylate L-asparagine at ~10-fold lower maximal velocity, underscoring the strict optimization of the active site for L-aspartate (PMID: 18004787).

The catalytic mechanism

The reaction follows an **ordered bi-bi kinetic mechanism**:

Step	Event
1	Carbamoyl phosphate (CP) binds first to the CP-binding domain; Arg54 (E. coli numbering) contacts the anhydride and phosphate oxygens
2	CP binding organizes the active site and protects the labile intermediate from thermal decomposition
3	L-aspartate binds to the adjacent aspartate-binding domain; domain closure creates the high-activity active site
4	The α-amino group of aspartate performs nucleophilic attack on the CP carbonyl carbon → tetrahedral intermediate
5	Intramolecular proton transfer; collapse of intermediate → N-carbamoyl-L-aspartate + Pi
6	Products released

The active site is **shared between two adjacent catalytic chains** — residues from one chain (e.g., Arg54, Arg105, Leu267) and from the neighboring chain (e.g., Lys84) both contribute — which is why the trimeric organization of catalytic subunits is essential for activity.

The distinctive *Pseudomonas*-type quaternary structure

The single most important organism-specific insight is the enzyme's **architecture**:

Feature	*E. coli* ATCase (class C)	*P. putida* ATCase (class A / *Pseudomonas*-type)
Catalytic subunit	PyrB (~34 kDa), 6 copies	PyrB (36.4 kDa, 334 aa), 6 copies
Regulatory subunit	PyrI (~17 kDa), 6 copies	**None** — no *pyrI* gene in genome
Second subunit type	—	Inactive DHOase homolog **PyrC′** (44.2 kDa), 6 copies
Holoenzyme	Heterododecamer $2C_3{:}3R_2$	**Dodecamer $PyrB_6/PyrC'_6$**
Regulatory nucleotide site	On separate PyrI chains	In **N-terminal extension of PyrB itself**
Role of second subunit	Allosteric regulation	**Structural — maintains dodecameric assembly**

In *P. putida*, the regulatory function has been internalized into an N-terminal extension of PyrB (explaining the ~23-residue length increase over *E. coli* PyrB and the C-terminal shift of the catalytic motifs in the sequence). The PyrC′ protein — a "vestigial" dihydroorotase that has lost its catalytic residues — has been co-opted as an obligatory structural scaffold. This is an elegant example of **enzyme recruitment / moonlighting**, in which an ancestral catalytic protein is retained for a purely architectural purpose. The genome-level confirmation that a *separate*, genuine *pyrC* dihydroorotase (PP_1086) exists elsewhere resolves any ambiguity: the pathway's DHOase chemistry is provided by PP_1086, while PP_4999 (PyrC′) exists only to build the ATCase holoenzyme.

Regulation

The enzyme is controlled at two levels: 1. **Allosteric (post-translational):** via the nucleotide-binding regulatory site in the PyrB N-terminal extension, tuning activity in response to the nucleotide pool. 2. **Transcriptional (gene expression):** pyrimidine end-product (uracil/UMP-derived) repression of the *pyr* regulon, as demonstrated for the closely grouped *Thermus* and *Pseudomonas* systems (PMID: 9171389).

Evidence Base

PMID	Title (abbreviated)	Role in this report
7896697	*Aspartate transcarbamoylase genes of Pseudomonas putida: requirement for an inactive dihydroorotase for assembly into the dodecameric holoenzyme*	**Primary, organism-specific evidence.** Defines the 334-aa/36.4-kDa PyrB catalytic subunit, the inactive PyrC′ DHOase homolog, the dodecameric $PyrB_6/PyrC'_6$ assembly, the internal N-terminal regulatory site, and the absence of dissociable regulatory subunits.
1633169	*13C and 15N isotope effects as a probe of the chemical mechanism of E. coli aspartate transcarbamylase*	Defines the chemical mechanism: nucleophilic attack, tetrahedral intermediate, product formation.
18053734	*Expression and functional analysis of ATCase … in Arabidopsis*	Establishes ATCase as the committed step of de novo UMP biosynthesis.
18971327	*Mechanism of thermal decomposition of carbamoyl phosphate and its stabilization by aspartate and ornithine transcarbamoylases*	Establishes the ordered mechanism (CP binds first) and intermediate stabilization.
1303763	*Arginine 54 in the active site of E. coli aspartate transcarbamoylase is critical for catalysis*	Identifies the conserved catalytic CP-binding arginine (Arg54), used to anchor the sequence-motif conservation in Q88D30.
17603076	*Structural model of the R state of E. coli ATCase with substrates bound*	Documents active-site residues (Arg105, Leu267) and pre-catalytic geometry.
10386880	*The 80s loop … is critical for catalysis and homotropic cooperativity*	Documents the cross-subunit Lys84 contribution to the shared active site.
9171389	*Structure and expression of a pyrimidine gene cluster from Thermus strain ZO5*	Confirms *Pseudomonas* ATCase sequence grouping (~50% identity) and pyrimidine transcriptional repression of *pyrB*.
7479879	*Crystal structure of P. aeruginosa catabolic OTCase … different oligomeric organization*	Illustrates the structural diversity of *Pseudomonas* transcarbamoylases (dodecameric assemblies).
18004787		

PMID	Title (abbreviated)	Role in this report
	Use of L-asparagine … to investigate catalysis and cooperativity in E. coli ATCase	Demonstrates strict substrate specificity for L-aspartate (asparagine carbamoylated ~10× slower).

The strongest and most directly relevant paper is **Schurr et al. 1995 (PMID: 7896697)**, which characterized the *P. putida* enzyme itself and provides organism-specific experimental grounding for essentially all of the structural conclusions. The mechanistic and active-site details are transferred from the exhaustively studied *E. coli* ortholog, which is justified by the ~39% sequence identity and exact conservation of the catalytic motifs.

Limitations and Knowledge Gaps

1. **No direct structural or kinetic study of Q88D30 itself.** The mechanistic and quaternary-structure conclusions rest on (a) the *P. putida* enzyme characterized by Schurr et al. and (b) the *E. coli* ortholog. No crystal structure, cryo-EM structure, or steady-state kinetic characterization of the specific KT2440 gene product (PP_4998) has been identified. Confirmation that KT2440's enzyme behaves identically to the strain studied by Schurr et al. is by inference from genome conservation, not direct measurement.
2. **Regulatory ligand specificity is not resolved for KT2440.** While the regulatory site is placed in the PyrB N-terminal extension, the identity of the physiological allosteric effector(s) (e.g., ATP, CTP, UTP) and their quantitative effects on the KT2440 enzyme have not been experimentally determined here.
3. **The exact residues of the N-terminal regulatory site are not mapped.** Sequence analysis localizes the extension but does not define which residues bind nucleotides.
4. **Transcriptional regulation of the *P. putida pyr* regulon is inferred by analogy.** Pyrimidine end-product repression of *pyrB* is documented in *Thermus* (and is a general feature of *pyr* regulons); direct promoter/operator mapping for PP_4998 in KT2440 was not performed.
5. **The UniProt SUBUNIT annotation conflict** (rule-based "PyrI heterododecamer") is a documented database inaccuracy for this organism rather than a genuine biological uncertainty, but downstream users relying solely on automated annotation may be misled.

Proposed Follow-up Experiments and Actions

1. **Recombinant expression and enzymology of PP_4998.** Clone and overexpress KT2440 *pyrB* (with and without co-expressed PP_4999/PyrC′), purify, and determine steady-state kinetic parameters (kcat, Km for CP and L-aspartate) and substrate specificity. Confirm that co-expression of PyrC′ is required for a stable, active dodecamer.
2. **Structural determination.** Solve the cryo-EM or crystal structure of the KT2440 $PyrB_6$/$PyrC'_6$ holoenzyme to directly visualize the dodecameric assembly, the PyrB–PyrC′ interfaces, and the N-terminal regulatory extension.
3. **Map the allosteric site.** Perform nucleotide-binding assays (ITC/fluorescence) with ATP, CTP, and UTP against the holoenzyme, and use mutagenesis of the PyrB N-terminal extension to identify effector-binding residues.
4. **Genetic validation in KT2440.** Construct *pyrB* (PP_4998) and *pyrC′* (PP_4999) deletion mutants and test for pyrimidine auxotrophy and loss of ATCase activity, confirming the essential structural role of PyrC′ in vivo.
5. **Transcriptional analysis.** Measure *pyrBC′* transcript levels under pyrimidine-rich versus pyrimidine-limited conditions (qRT-PCR/RNA-seq) to confirm end-product repression of the *P. putida pyr* operon.
6. **Correct the annotation.** Submit a curation note to UniProt flagging that the rule-based SUBUNIT field (PyrI-containing heterododecamer) is inaccurate for *P. putida*, which uses the *Pseudomonas*-type $PyrB_6$/$PyrC'_6$ architecture with no PyrI.

Conclusion

pyrB (PP_4998, Q88D30) is unambiguously the catalytic subunit of aspartate carbamoyltransferase (EC 2.1.3.2) in *Pseudomonas putida* KT2440. It catalyzes the first committed, cytoplasmic step of de novo pyrimidine biosynthesis — carbamoylation of L-aspartate by carbamoyl phosphate to form N-carbamoyl-L-aspartate — via an ordered mechanism with high specificity for L-aspartate. The physiological enzyme is a *Pseudomonas*-type dodecamer of six catalytic PyrB subunits and six catalytically inactive dihydroorotase-homolog PyrC′ subunits, with the allosteric regulatory site integrated into an N-terminal extension of PyrB; there is no separate PyrI regulatory subunit, making the widely propagated *E. coli*-style subunit annotation incorrect for this organism.